TOPIC 5 · HYDROSTATICS

Solved Problems: Official Hydrostatics Sheets FS1 to FS10

Worked solutions to the hydrostatics sheets of the course (Procesos Fluidotérmicos, Escuela Politécnica Superior). Each statement is translated from the Spanish original and redrawn; sheet FS8 is a second scan of FS2 and is solved there. Every solution follows the order used in class: hypotheses and data; the coordinates, the surface element and its normal; the pressure field (5.10) and the forces and moments (5.11)-(5.16), integrated step by step and checked with the projection rule or Archimedes' principle; the answer; and the checks (limiting cases, dimensions, official key). Three problems carry an interactive simulator inside their solution: the plane gate of FS4, the cylinder of FS10 and the floating cube of FS3. References to Notes.pdf are to Chapter 5 (Hydrostatics).

PROBLEM FS1

Melting Ice Cube

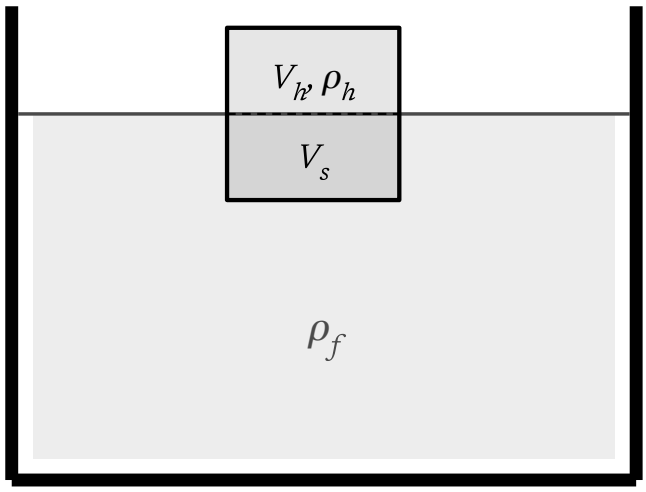
Original statement · fluid_statics_1.pdf

Archimedes | Floating body | Mass conservation

OFFICIAL PROBLEM STATEMENT

An ice cube of density ρ_h is partially submerged in a container holding a fluid of density $\rho_f > \rho_h$. The total volume of the ice cube is V_h, while the submerged part of the cube occupies a volume V_s to be determined. Calculate the volume of water displaced once the cube has melted, turning into water of density ρ_a, as a function of ρ_f, ρ_h, ρ_a and V_h, knowing that $\rho_h < \rho_a < \rho_f$. Particularize for the case $\rho_f = \rho_a$.

Translated from the Spanish original; the figure of the sheet labels the cube density ρ_b, the same quantity as ρ_h.



ice cube: $\rho_h < \rho_f$

melt water: ρ_a

$\rho_h < \rho_a < \rho_f$

Statement figure (redrawn from fluid_statics_1.pdf, p. 1): ice cube of volume V_h and density ρ_h floating in a liquid of density $\rho_f > \rho_h$, submerged volume V_s

`Source: fluid_statics_1.pdf (page 1, sheet of 11-2-00). Notes.pdf, Chapter 5: Archimedes' principle, Eq. 5.15.`

Worked solution

Hypotheses, Data and Unknowns

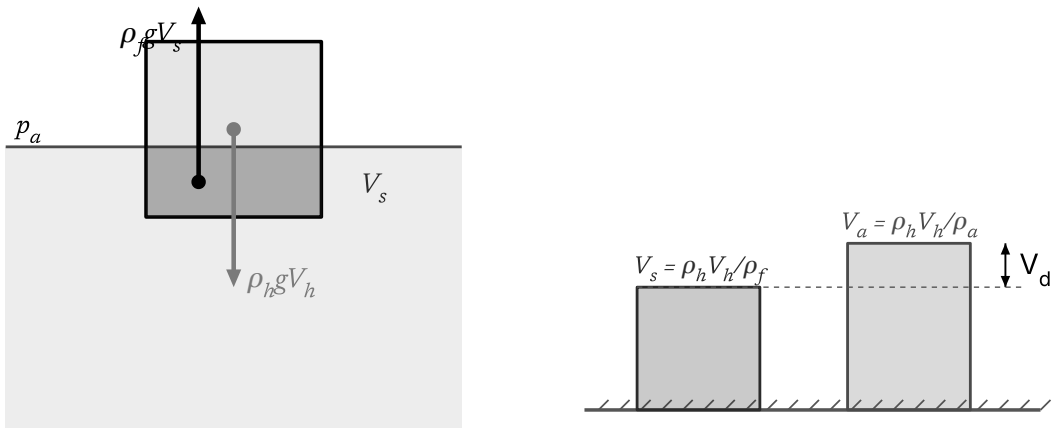
- The fluid and the melt water are at rest and incompressible; the air above them is at the uniform pressure p_a.
- The cube floats in equilibrium with part of its volume above the free surface (this requires $\rho_h < \rho_f$, given).
- Mass is conserved when the ice melts; the melt water and the fluid do not change volume when they meet (the melt water either floats as a layer, since $\rho_a < \rho_f$, or mixes with additive volumes).
- Data: $\rho_h, \rho_f, \rho_a, V_h, g$. Unknowns: the submerged volume V_s and the volume V_d that the liquid gains when the cube melts, that is the change of the volume of liquid below the free surface.

Symbol	Meaning	SI unit
ρ_h, ρ_f, ρ_a	densities of ice, fluid and melt water	kg/m^3

Symbol	Meaning	SI unit
V_h	volume of the cube	m^3
V_s	submerged volume of the cube	m^3
V_a	volume of the melt water	m^3

The Floating Cube: Archimedes' Principle

The forces on the cube are its weight and the pressure of the fluid and of the air on its surface. The air pressure p_a acts over the whole closed surface of the cube and adds up to zero ($\oint p_a \vec{n}\, d\sigma = 0$). The part below the free surface feels the gauge pressure of the fluid, and by Archimedes' principle (Eq. 5.15) applied to the submerged part, closed by the plane of the free surface where the gauge pressure is zero, its resultant is the weight of the displaced fluid, $\rho_f g V_s$, upward.



(a) floating cube: weight and buoyancy

(b) after melting: $V_d = V_a - V_s$

(a) The floating cube: weight $\rho_h g V_h$ at its centre of gravity, buoyancy $\rho_f g V_s$ at the centroid of the submerged part. (b) The cube displaced V_s; once melted, the same mass occupies $V_a = \rho_h V_h/\rho_a$; the difference $V_d = V_a - V_s$ is the volume the liquid gains

Vertical equilibrium of the cube (upward positive):

$$\rho_f g V_s - \rho_h g V_h = 0 \quad \Longrightarrow \quad V_s = \frac{\rho_h}{\rho_f} V_h$$

Since $\rho_h < \rho_f$, $V_s < V_h$: part of the cube emerges, as assumed.

Melting: Conservation of Mass

When the ice melts, its mass $\rho_h V_h$ is conserved and becomes water of density ρ_a:

$$\rho_a V_a = \rho_h V_h \quad \Longrightarrow \quad V_a = \frac{\rho_h}{\rho_a}\, V_h$$

Before melting, the cube occupied the volume V_s below the free surface; after melting, the melt water occupies V_a in the liquid. The volume of liquid below the free surface therefore grows by

$$V_d = V_a - V_s = \rho_h V_h \left(\frac{1}{\rho_a} - \frac{1}{\rho_f}\right) = \frac{\rho_h}{\rho_a}\, V_h \left(1 - \frac{\rho_a}{\rho_f}\right)$$

In a container of horizontal section S, the free surface rises by V_d / S.

Answers and Checks

- Submerged volume: $V_s = (\rho_h / \rho_f)\, V_h$.
- Extra volume after melting: $V_d = (\rho_h / \rho_a)\, V_h\, (1 - \rho_a / \rho_f)$. With $\rho_a < \rho_f$ it is positive: the level rises, because the melt water is lighter than the fluid it displaced.
- Particular case $\rho_f = \rho_a$ (ice floating in its own water): $V_d = 0$. The melt water fills exactly the hole the cube had made, and the level does not change.
- Limiting case $\rho_h \to \rho_f$: $V_s \to V_h$, the cube is just fully submerged.
- Dimensions: V_d is a ratio of densities times V_h, so m^3; both brackets are dimensionless.
- Check against the official key: fluid_statics_1.pdf gives $V_s = (\rho_h/\rho_f)V_h$ and $V_d = (\rho_h/\rho_a)V_h(1 - \rho_a/\rho_f)$, the same results.

PROBLEM FS2

Hemispherical Station on the Sea Floor

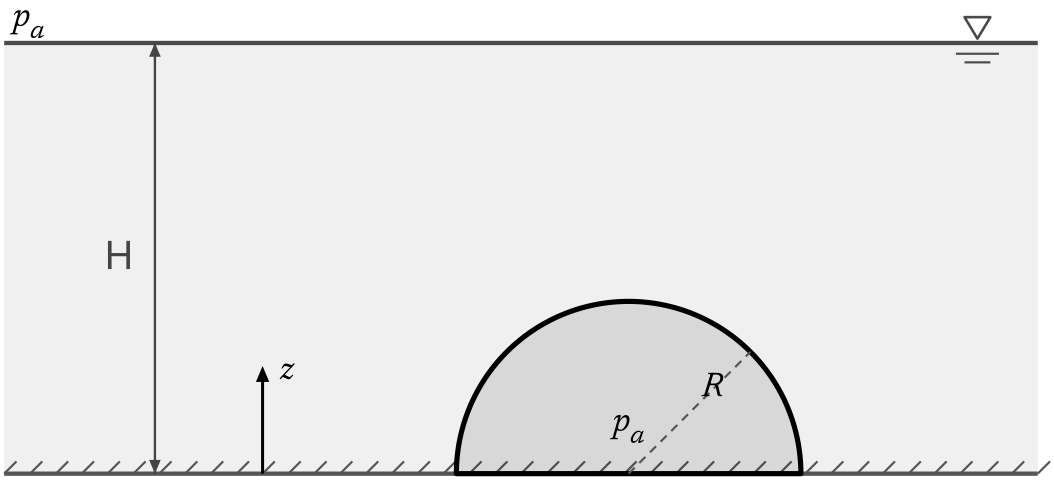
Original statement · fluid_statics_2.pdf

Curved surface | Spherical coordinates | Projection rule | Sheets FS2 and FS8

OFFICIAL PROBLEM STATEMENT

A hemispherical experimental station of radius R is to be installed on the ocean floor at a depth H. Calculate the force, indicating its direction and sense, exerted on the walls of the station, knowing that the pressure inside is equal to the atmospheric pressure, p_a.

Translated from the Spanish original.



Statement figure (redrawn from fluid_statics_2.pdf and fluid_statics_8.pdf, p. 1): hemispherical station of radius R on the sea floor at depth H, atmospheric pressure p_a inside, axis z up from the floor

Source: fluid_statics_2.pdf and fluid_statics_8.pdf (page 1, sheet of 29-01-01). Notes.pdf, Chapter 5: Eqs. 5.10, 5.11 and 5.14.

Worked solution

Hypotheses, Data and Unknowns

- Sea water at rest, incompressible, density ρ; the free surface is at the atmospheric pressure p_a, the depth of the floor is $H > R$.
- The walls form a thin hemispherical shell of radius R resting on the floor; inside, the air is at p_a. The force asked for is the net pressure force on the shell (water outside plus air inside); the floor of the station is not part of the walls.
- Data: R, H, ρ, g, p_a. Unknown: the force $\vec{F}$.

Symbol	Meaning	SI unit
R	radius of the station	m
H	depth of the sea floor	m
ρ	density of sea water	kg/m^3

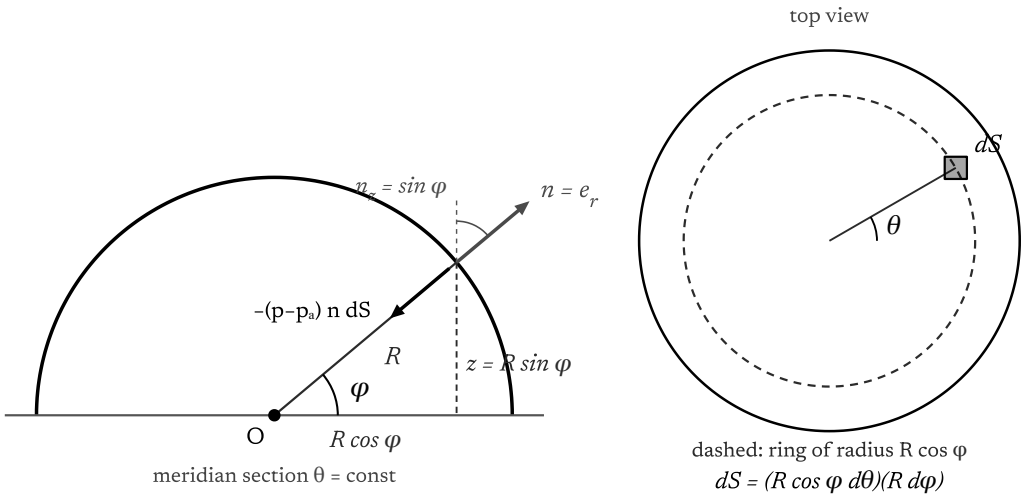
Coordinates, Surface Element and Normal

Take the origin O at the centre of the base of the hemisphere, z vertical and pointing up, so the floor is $z = 0$ and the free surface $z = H$. A point of the shell is located by its **elevation** φ above the floor plane ($0 \le \varphi \le \pi/2$) and its **azimuth** θ ($0 \le \theta < 2\pi$). In the meridian section (right triangle with the radius R as hypotenuse), the height $z = R\sin\varphi$ is the leg opposite to φ and the distance to the axis $R\cos\varphi$ the adjacent leg.

The element of the sphere is a small rectangle with sides $R\,d\varphi$ (along the meridian) and $R\cos\varphi\,d\theta$ (along the parallel, a circle of radius $R\cos\varphi$):

$$dS = R^2\cos\varphi\,d\theta\,d\varphi$$

The unit normal towards the water is the radial vector $\vec{e}_r = \cos\varphi\cos\theta\,\vec{e}_x + \cos\varphi\sin\theta\,\vec{e}_y + \sin\varphi\,\vec{e}_z$; its vertical component $n_z = \sin\varphi$ is the leg opposite to φ of the unit right triangle.



Spherical coordinates of the station. Left: meridian section; the radius R is the hypotenuse, $z = R\sin\varphi$ is the leg opposite to the elevation φ and $R\cos\varphi$ the adjacent leg; the outward normal is $\vec{e}_r$, with $n_z = \sin\varphi$; the water pushes along $-\vec{e}_r$. Right: top view; at elevation φ the parallel is a ring of radius $R\cos\varphi$, so the element is $dS = (R\cos\varphi\, d\theta)(R\, d\varphi)$

The Pressure and the Net Elementary Force

By (5.10), at the height z (depth $H - z$): $p = p_a + \rho g(H - z) = p_a + \rho g(H - R\sin\varphi)$.

The water pushes the outer face with $-p\,\vec{e}_r\,dS$; the air pushes the inner face with $-p_a(-\vec{e}_r)\,dS$, since the normal towards the air is $-\vec{e}_r$. Their sum is

$$d\vec{F} = -(p - p_a)\,\vec{e}_r\,dS = -\rho g(H - R\sin\varphi)\,\vec{e}_r\,R^2\cos\varphi\,d\theta\,d\varphi$$

Integration

Horizontal components. F_x contains $\int_0^{2\pi}\cos\theta\,d\theta = [\sin\theta]_0^{2\pi} = 0$, and F_y contains $\int_0^{2\pi}\sin\theta\,d\theta = [-\cos\theta]_0^{2\pi} = 0$: by the axial symmetry the horizontal forces cancel.

Vertical component. With $n_z = \sin\varphi$, the θ integral of a function independent of θ gives 2π:

$$F_z = -\int_0^{\pi/2}\int_0^{2\pi} \rho g(H - R\sin\varphi)\sin\varphi\, R^2\cos\varphi\, d\theta\, d\varphi = -2\pi R^2\rho g \int_0^{\pi/2} \left(H\sin\varphi\cos\varphi - R\sin^2\varphi\cos\varphi\right) d\varphi$$

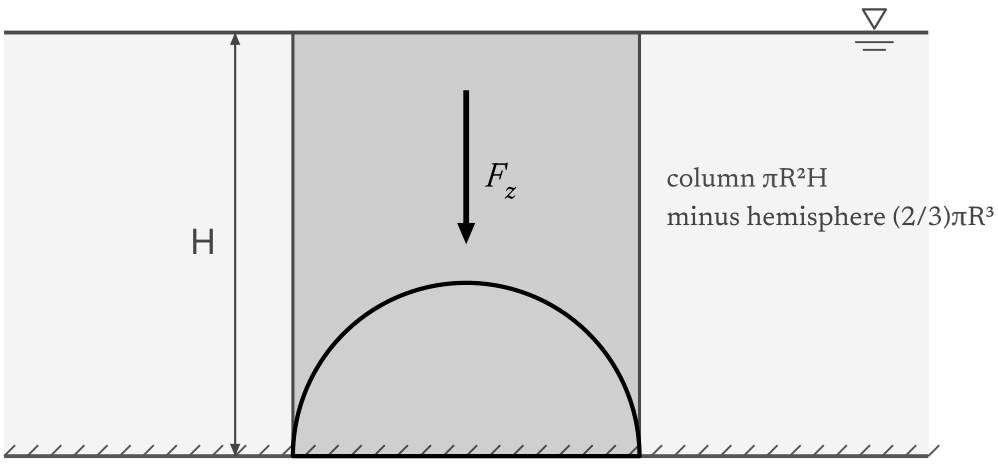
Both terms are integrated with the substitution $u = \sin\varphi, du = \cos\varphi\, d\varphi$, with $u = 0$ at $\varphi = 0$ and $u = 1$ at $\varphi = \pi/2$:

$$\int_0^{\pi/2} \sin\varphi\cos\varphi\, d\varphi = \int_0^1 u\, du = \left[\frac{u^2}{2}\right]_0^1 = \frac{1}{2}, \qquad \int_0^{\pi/2} \sin^2\varphi\cos\varphi\, d\varphi = \int_0^1 u^2\, du = \left[\frac{u^3}{3}\right]_0^1 = \frac{1}{3}$$

Therefore

$$F_z = -2\pi R^2\rho g\left(\frac{H}{2} - \frac{R}{3}\right) = -\pi R^2\rho g\left(H - \frac{2R}{3}\right)$$

Check with the Projection Rule



Projection rule (Eq. 5.14): F_z is the weight of the water that would stand above the hemisphere, the cylinder $\pi R^2 H$ minus the hemisphere $\frac{2}{3}\pi R^3$; it points down

By (5.14), the vertical force on a surface with the water above it is the weight of the water column standing on it, down to the surface: the cylinder of radius R and height H, $\pi R^2 H$, minus the volume of the hemisphere, $\frac{2}{3}\pi R^3$, which is not water:

$$F_z = -\rho g \left(\pi R^2 H - \frac{2}{3}\pi R^3 \right) = -\pi R^2 \rho g \left(H - \frac{2R}{3} \right)$$

the same result. The air inside contributes nothing net because p_a also acts at the top of the column.

Answers and Checks

- $\vec{F} = -\pi R^2 \rho g \, (H - 2R/3) \, \vec{e}_z$: vertical, downward (since $H > R > 2R/3$), pressing the station against the floor; no horizontal component.
- **Limiting case** of a small station, $R \ll H$: $F_z \approx -\rho g H \, \pi R^2$, the gauge pressure at the floor times the base area, as for a flat lid; the term $2R/3$ comes from the hemisphere volume, which is not water.
- **Physical reading**: the station is not lifted by Archimedes' principle, because no water reaches its base: the buoyancy formula (5.15) applies only to a surface closed by water.
- **Dimensions**: $\rho g R^2 H$ is (kg m^{-3})(m s^{-2})(m^3) = N.
- **Check against the official key**: fluid_statics_2.pdf (and its second scan fluid_statics_8.pdf) sets up the same integral, $F_z = -\iint \rho g (H - R\sin\varphi) \sin\varphi \, R^2 \cos\varphi \, d\theta \, d\varphi$, and reaches $-2\pi R^2 \rho g [H \sin^2\varphi/2 - R\sin^3\varphi/3]_0^{\pi/2}$, which equals the result above.

Discrepancy with the official solution

The handwritten last line reads $-\pi R^2 \rho g \, [H - 2R/3] \, \vec{e}_z$ preceded by a mark that looks like a struck-out 4 (reading of handwriting). The previous line of the key, $-2\pi R^2 \rho g \, [H \sin^2\varphi/2 - R\sin^3\varphi/3]_0^{\pi/2}$, evaluates to $-2\pi R^2 \rho g \, (H/2 - R/3) = -\pi R^2 \rho g \, (H - 2R/3)$, coefficient π, as above.

PROBLEM FS3

Hollow Cube with Trapped Air

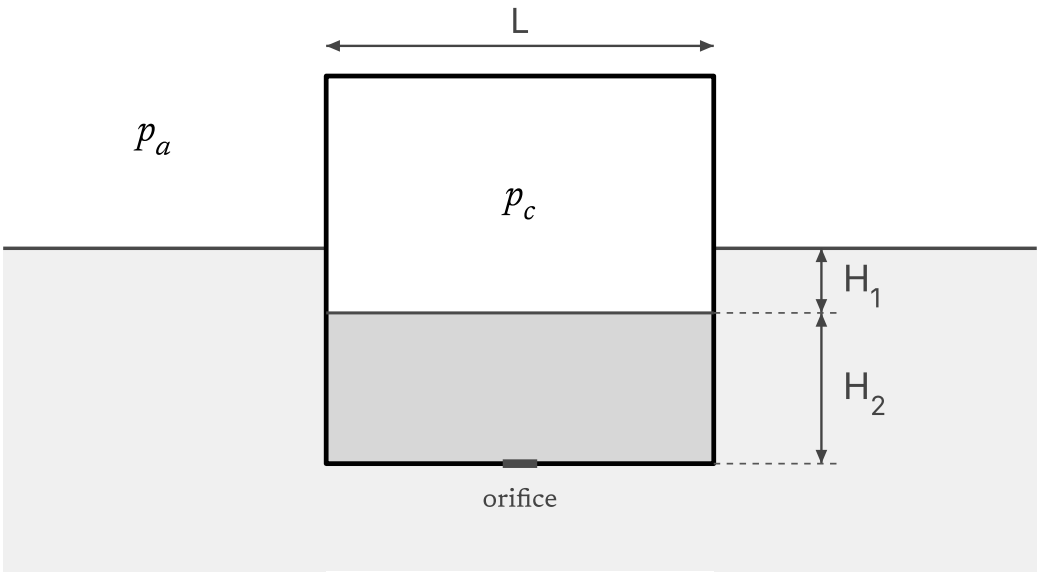
Original statement · fluid_statics_3.pdf

Floating body | Trapped air | Isothermal gas | Simulator

OFFICIAL PROBLEM STATEMENT

A hollow cube of side L and weight $W < \rho g L^3$ is initially full of air at the ambient pressure p_a and temperature T_a. At a certain instant, the cube is placed on the surface of a liquid of density ρ. The liquid enters the cube through an orifice in its base until an equilibrium position is reached, characterized by the lengths H_1 and H_2 of the attached figure. Assuming that the evolution of the air inside the cube is isothermal, determine the values of H_1/L and H_2/L, as well as the final pressure inside the cube, p_c.

Translated from the Spanish original.



Statement figure (redrawn from fluid_statics_3.pdf, p. 1): hollow cube of side L with an orifice in its base; the inner interface is H_1 below the outer free surface and H_2 above the base; air at p_c inside, p_a outside

Source: fluid_statics_3.pdf (page 1, sheet of 27-01-03). Notes.pdf, Chapter 5: Eqs. 5.10, 5.15 and 5.18.

Worked solution

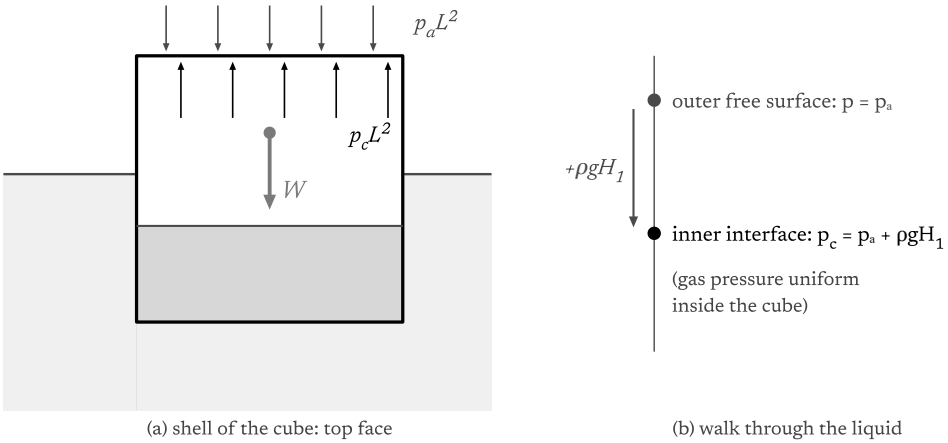
Hypotheses, Data and Unknowns

- Liquid and air at rest; the liquid is incompressible; the outside air is at the uniform pressure p_a.
- The trapped air has a uniform pressure p_c (its weight and the change of its pressure over a height L are negligible, Section 7 of Topic 5) and keeps the temperature T_a: isothermal process, $p\,V = \text{const}$ from the state equation (5.18) at constant T and constant mass.
- Thin walls: the volume of the walls is neglected; W is the weight of the cube alone (the air weight is negligible).
- The cube floats upright with its top face above the outer free surface (checked at the end).
- Data: L, W, ρ, g, p_a. Unknowns: H_1 (depth of the inner interface below the outer free surface), H_2 (height of the inner liquid layer above the base) and p_c.

Symbol	Meaning	SI unit
L	side of the cube	m
W	weight of the cube	N
H_1, H_2	lengths of the figure	m
p_a, p_c	outside pressure, final air pressure inside	Pa

Coordinates and the Free-Body Diagram

Take the vertical axis pointing up. The system for the force balance is the **shell of the cube** alone (not the air, not the liquid inside). Its side walls are vertical, so the pressure on them is horizontal and adds up to zero by symmetry; its base has an orifice and is wetted by the same liquid on both faces at the same depth, so its net pressure force is zero (thin wall). Only the top face carries a net vertical pressure force: $p_a L^2$ downward from the outside air and $p_c L^2$ upward from the trapped air.



(a) Vertical forces on the shell of the cube (thin walls): p_aL^2 down on the top face from outside, p_cL^2 up on it from the trapped air, weight W down; the side walls carry only horizontal forces, which cancel. (b) Walking from the outer free surface down the depth H_1 to the inner interface gives $p_c = p_a + \rho g H_1$

Equilibrium of the Shell

Vertical equilibrium of the shell (upward positive):

$$p_cL^2 - p_aL^2 - W = 0 \quad \Longrightarrow \quad p_c = p_a + \frac{W}{L^2}$$

The Hydrostatic Law between the Two Free Surfaces

The liquid inside the cube is connected with the outside liquid through the orifice, so (5.10) holds through both. Walking from a point of the outer free surface, where $p = p_a$, down a depth H_1 to the level of the inner interface (any path inside the connected liquid gives the same result), the pressure grows by $\rho g H_1$; at the interface it equals the air pressure (Section 1 of Topic 5, no surface tension):

$$p_c = p_a + \rho g H_1$$

Equating the two expressions of p_c:

$$\rho g H_1 = \frac{W}{L^2} \quad \Longrightarrow \quad \frac{H_1}{L} = \frac{W}{\rho g L^3}$$

The hypothesis $W < \rho g L^3$ gives $H_1 < L$.

The Trapped Air: Isothermal Compression

Initially the air fills the whole cube, $V_0 = L^3$, at p_a. At equilibrium it fills the part of the cube above the inner interface, a prism of base L^2 and height $L - H_2$, at p_c. With the same mass and the same temperature, $p\,V$ is constant:

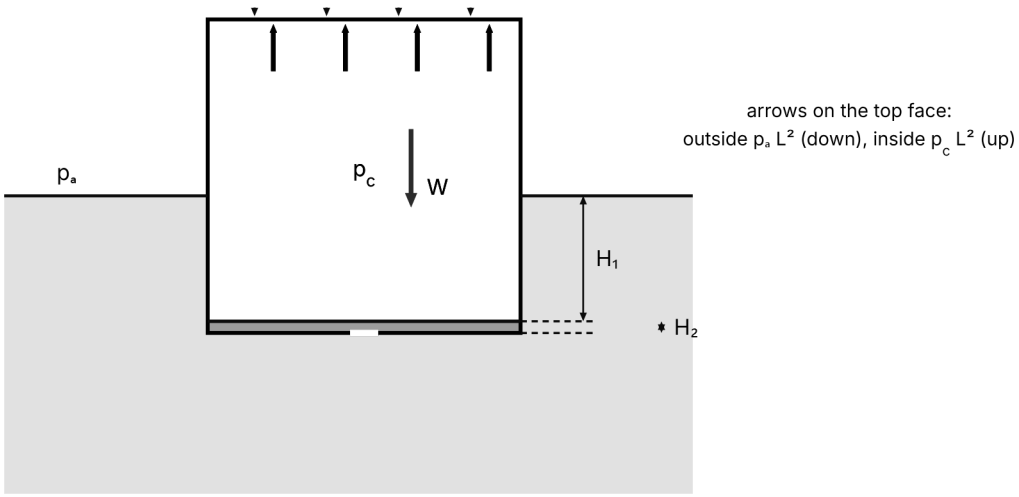
$$p_a L^3 = p_c L^2 (L - H_2) \quad \Longrightarrow \quad \frac{L - H_2}{L} = \frac{p_a}{p_c} \quad \Longrightarrow \quad \frac{H_2}{L} = 1 - \frac{p_a}{p_c} = \frac{p_c - p_a}{p_c}$$

and with $p_c - p_a = W/L^2$ and $p_c = p_a + W/L^2$, multiplying numerator and denominator by L^2:

$$\frac{H_2}{L} = \frac{W/L^2}{p_a + W/L^2} = \frac{W}{p_a L^2 + W}$$

Simulator: the Floating Cube

Simulator: floating hollow cube with trapped air



$H_1/L = W/(\rho g L^3)$	0.4000
$H_2/L = W/(p_a L^2 + W)$	0.0374
$p_c/p_a = 1 + W/(p_a L^2)$	1.0388
Check $(p_c - p_a)L^2 = W$	both equal 0.0388 $p_a L^2$
Top face above the outer surface?	yes: $H_1 + H_2 = 0.437\ L < L$

Answers and Checks

- $H_1/L = W/(\rho g L^3)$, $H_2/L = W/(p_a L^2 + W)$, $p_c = p_a + W/L^2$.
- **Archimedes check.** For the system cube plus trapped air, the displaced liquid is the air below the outer free surface, a prism of base L^2 and height H_1 (from the outer level down to the inner interface). Its weight $\rho g L^2 H_1 = W$ balances the cube, which is the same equation as above.
- **Limiting cases.** $W \to 0$: $H_1 \to 0$, $H_2 \to 0$, $p_c \to p_a$ (a weightless box lets no liquid in). $p_a \to \infty$ (incompressible air): $H_2 \to 0$, the air keeps its volume.
- **Consistency.** The top face is above the outer free surface when $H_1 + H_2 < L$, that is $W/(\rho g L^3) + W/(p_a L^2 + W) < 1$. For water and $L = 1$ m, $p_a/(\rho g L) \approx 10.3$, so $H_2 \ll H_1$: the air is barely compressed and the cube floats almost as a closed box.
- **Dimensions.** $W/(\rho g L^3)$: N / (kg m^{-3} m s^{-2} m^3) = N/N, dimensionless; $W/(p_a L^2)$: N/(Pa m^2) = N/N; W/L^2 in Pa.

- **Check against the official key.** fluid_statics_3.pdf gives $H_1 = W/(\rho g L^2)$, $p_c = p_a + W/L^2$ and $H_2/L = 1 - p_a/p_c = W/(p_a L^2 + W)$, the same results.

PROBLEM FS4

Square Valve on its Centre Axis

Original statement · fluid_statics_4.pdf

Plane surface | Centre of pressure | Moment balance | Simulator

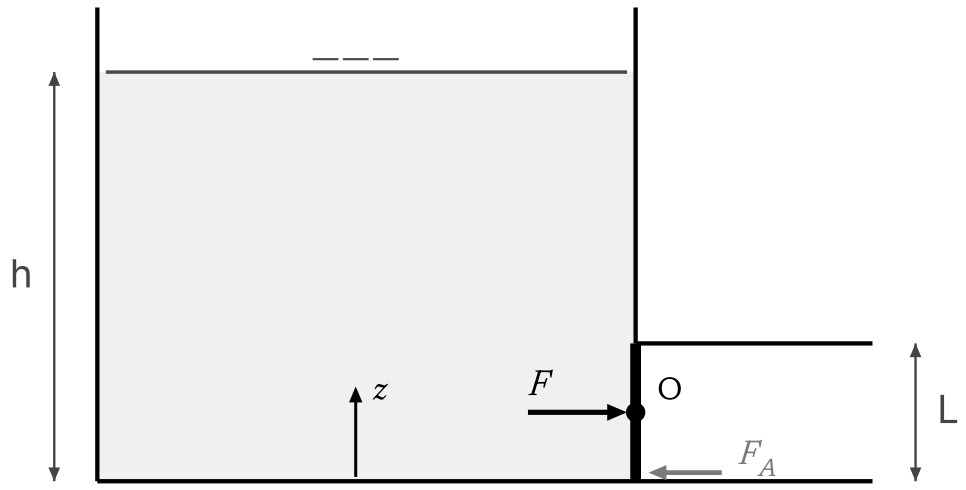
OFFICIAL PROBLEM STATEMENT

The square valve of side L shown in the figure can turn about the horizontal axis that passes through its centre. Determine:

(a) The force F that the fluid exerts on the valve.

(b) The value of the force F_A that we must apply at the base of the valve to keep it closed.

Translated from the Spanish original.



Statement figure (redrawn from fluid_statics_4.pdf, p. 1): square valve of side L at the bottom of the right wall, pivoting about the horizontal axis O through its centre; water depth h; F_A is applied at the base of the valve

Source: fluid_statics_4.pdf (page 1, sheet of 10-09-03). Notes.pdf, Chapter 5: Eqs. 5.10 and 5.11.

Worked solution

Hypotheses, Data and Unknowns

- Water at rest, incompressible, density ρ; air at p_a above the water and on the outer face of the valve, so only the gauge pressure counts (Section 4 of Topic 5).
- The valve is a vertical square plate of side L with its lower edge on the bottom of the tank and its top edge at the depth $h - L$ below the free surface ($h \geq L$).
- The axis is horizontal, in the plane of the valve, through its centre; friction on the axis is neglected; the weight of the valve acts through the axis (homogeneous plate) and has no moment about it.
- F_A is horizontal and acts at the base of the valve, as drawn.
- Data: L, h, ρ, g. Unknowns: F and F_A.

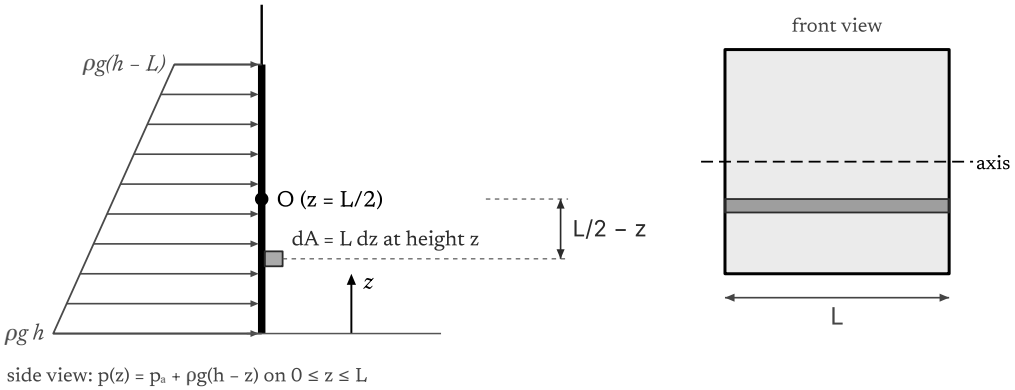
Symbol	Meaning	SI unit
L	side of the valve	m
h	depth of the water	m
F, F_A	water force, holding force	N

Coordinates and the Surface Element

Take z vertical, pointing up, with $z = 0$ at the bottom of the tank (the lower edge of the valve), so the valve covers $0 \leq z \leq L$ and the free surface is at $z = h$. The depth of a point is $h - z$, and (5.10) gives the gauge pressure

$$p(z) - p_a = \rho g (h - z)$$

The strip of the valve between z and $z + dz$ has the width L (horizontal side of the square) and the area $dA = L\,dz$. The unit normal towards the water is $\vec{n} = -\vec{e}_x$ (the water is on the left, x points to the right, out of the tank), so the elementary force $-(p - p_a)\vec{n}\,dA = (p - p_a)\,dA\,\vec{e}_x$ pushes the valve outward. The axis is at $z = L/2$; the strip is at the signed distance $L/2 - z$ below it.



Left: side view; the gauge pressure on the valve grows linearly from $\rho g(h-L)$ at the top edge to $\rho g h$ at the bottom; the strip $dA = L\,dz$ at height z has the arm $L/2 - z$ about the axis O. Right: front view of the valve, the horizontal axis through its centre and the strip

(a) The Force on the Valve

$$F = \int_0^L \rho g(h-z)\,L\,dz = \rho g L\left[hz - \frac{z^2}{2}\right]_0^L = \rho g L\left(hL - \frac{L^2}{2}\right) - 0 = \rho g L^2\left(h - \frac{L}{2}\right)$$

horizontal, pointing out of the tank ($+\vec{e}_x$). Check with the centroid (Section 4.1 of Topic 5): the centroid of the square is at the depth $h - L/2$, so $F = \rho g(h - L/2)L^2$, the same.

(b) The Moment about the Axis and the Holding Force

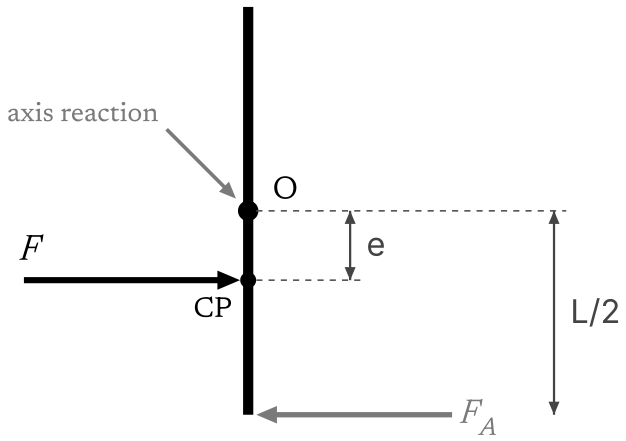
The strip below the axis ($z < L/2$) carries more pressure than the strip at the same distance above it, so the water tends to turn the valve with its lower edge outward. The moment about the axis, counted positive in that sense, is the force of each strip times its arm $L/2 - z$ (negative above the axis, where the strip turns the valve the other way):

$$M = \int_0^L \left(\frac{L}{2} - z\right)\rho g(h-z)\,L\,dz$$

Expanding the integrand, $(L/2 - z)(h - z) = \frac{Lh}{2} - z\left(\frac{L}{2} + h\right) + z^2$, and integrating term by term with Barrow's rule between 0 and L:

$$M = \rho g L \left[\frac{Lh}{2} z - \frac{z^2}{2} \left(\frac{L}{2} + h \right) + \frac{z^3}{3} \right]_0^L = \rho g L \left(\frac{L^2 h}{2} - \frac{L^3}{4} - \frac{L^2 h}{2} + \frac{L^3}{3} \right) = \rho g L \cdot \frac{L^3}{12} = \frac{\rho g L^4}{12}$$

The terms in h cancel: **the moment does not depend on the water depth**.



Free-body diagram of the valve: the water force F acts at the centre of pressure, a distance e below the axis O; F_A acts at the base with the arm $L/2$; the axis reaction passes through O and has no moment about it

Moment balance of the valve about its axis: the axis reaction and the weight pass through the axis; F_A, at the base, has the arm $L/2$ and must turn the valve the other way:

$$F_A \frac{L}{2} = \frac{\rho g L^4}{12} \quad \Longrightarrow \quad F_A = \frac{\rho g L^3}{6}$$

pointing into the tank ($-\vec{e}_x$).

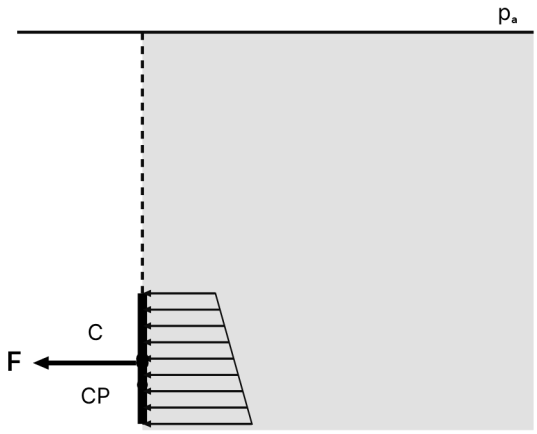
Check with the centre of pressure. The centre of pressure is below the centroid by $e = I_C/(s_C A)$ (Section 4.1, with $\theta = 90°$, s = depth): $I_C = L \cdot L^3/12$, $s_C = h - L/2$, $A = L^2$, so

$$e = \frac{L^4/12}{(h - L/2)L^2} = \frac{L^2}{12\,(h - L/2)}, \qquad F\,e = \rho g L^2 \left(h - \frac{L}{2}\right) \frac{L^2}{12\,(h - L/2)} = \frac{\rho g L^4}{12}$$

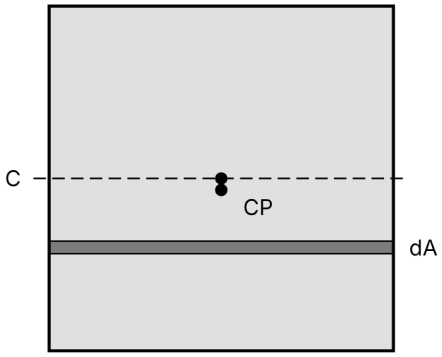
As the depth grows the centre of pressure approaches the centre ($e \to 0$) but the force grows in the same proportion, so the moment stays $\rho g I_C$ (the general result $\rho g \sin\theta\, I_C$ of Section 4.1).

Simulator: the Plane Gate

`Simulator: plane gate, pressure prism and centre of pressure`



side view



front view, scaled to fit the panel
C on the dashed axis, CP below it by e

Area A	`1.000 m²`
Centroid: s_C, depth	`2.500 m, 2.500 m`
F = ρg sinθ s_C A	`24.52 kN (normal to the gate)`
e = $s_{CP} - s_C = I_C/(s_C\,A)$	`33.3 mm`
Moment about the centroidal axis ρg sinθ I_C	`0.818 kN m (independent of h)`
Strip: s, gauge pressure	`2.700 m, 26.49 kPa`
Strip: w(s)	`1.000 m`
F_A at the lower edge (FS4, arm L/2)	`1.635 kN = ρgL³/6 when θ = 90°`

Answers and Checks

- (a) $F = \rho g L^2 (h - L/2)$, horizontal, out of the tank, at the depth $h - L/2 + L^2/(12(h - L/2))$.
- (b) $F_A = \rho g L^3/6$, horizontal, into the tank, independent of h.
- **Limiting case** $h = L$ (free surface at the top edge): $F = \rho g L^3/2$ and $e = L/6$, so the centre of pressure is at $L/3$ above the bottom, the centroid of the triangular pressure diagram.
- **Dimensions**: $\rho g L^3$ is (kg m^{-3})(m s^{-2})(m^3) = N.
- **Check against the official key**: fluid_statics_4.pdf gives $F = \rho g L^2 (h - L/2)$, $M = \rho g L^4/12$ and $F_A = \rho g L^3/6$, the same results.

PROBLEM FS5

Gate between Two Liquids

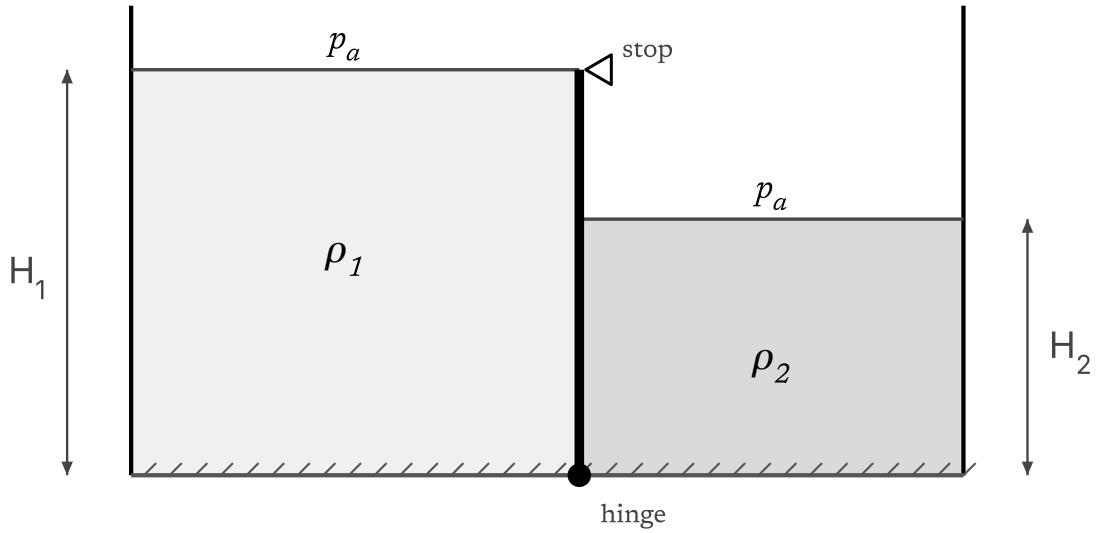
Original statement · fluid_statics_5.pdf

Plane surface | Two liquids | Moment about a hinge | Critical level

OFFICIAL PROBLEM STATEMENT

The gate of the attached figure separates two liquids of densities ρ_1 and $\rho_2 > \rho_1$. The height of the liquid on the two sides of the gate is H_1 and $H_2 < H_1$. The gate is hinged at its lower end and rests on a stop at its upper end, as shown in the attached figure, so that the configuration is in equilibrium as long as H_2 stays below a certain critical value, H_c, to be determined as a function of H_1, ρ_1 and ρ_2.

Translated from the Spanish original. The sheet gives no solution for this problem.



Statement figure (redrawn from fluid_statics_5.pdf, p. 1): gate of height H_1 between the liquids ρ_1 (left, height H_1) and $\rho_2 > \rho_1$ (right, height $H_2 < H_1$), hinged at the bottom and resting on a stop at the top

Source: fluid_statics_5.pdf (page 1, sheet of 11-2-00). Notes.pdf, Chapter 5: Eqs. 5.10 and 5.11.

Worked solution

Hypotheses, Data and Unknowns

- Both liquids at rest, incompressible, immiscible and separated by the gate; air at p_a above both, so only gauge pressures count.
- The gate is a vertical rigid plate of height H_1 and width b (normal to the paper), hinged at the bottom (point O) without friction; its weight passes through the hinge when it is vertical and has no moment about O.
- The stop at the top is on the side of liquid 2: it can only push the gate towards liquid 1, never pull.
- Data: ρ_1, ρ_2, H_1, H_2, g, b. Unknowns: the critical level H_c (and, as a by-product, the force on the stop).

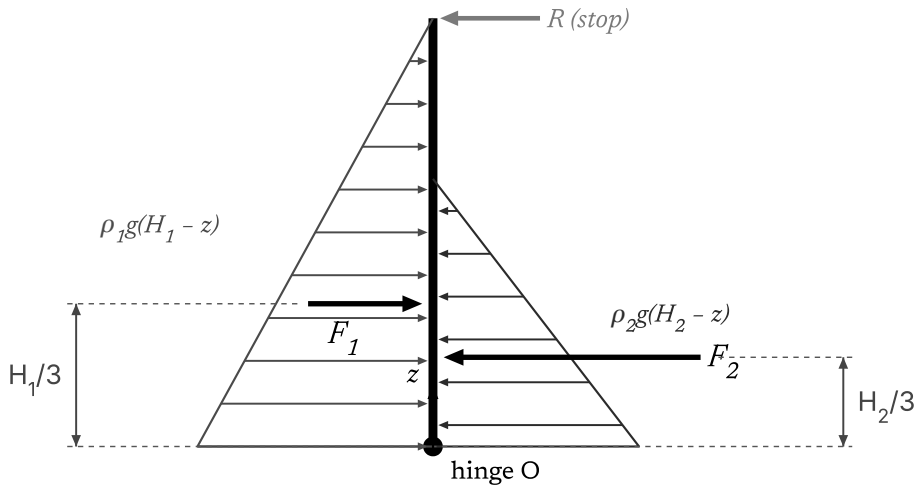
Symbol	Meaning	SI unit
ρ_1, ρ_2	densities, $\rho_2 > \rho_1$	kg/m^3
H_1, H_2	liquid heights	m
b	width of the gate	m

Coordinates and the Pressure on Each Face

Take z vertical, pointing up from the hinge, so the gate covers $0 \le z \le H_1$. By (5.10), measured from each free surface:

$$p_1(z) - p_a = \rho_1 g (H_1 - z), \quad 0 \le z \le H_1; \qquad p_2(z) - p_a = \rho_2 g (H_2 - z), \quad 0 \le z \le H_2$$

The strip between z and $z + dz$ has the area $b\,dz$, its force is horizontal, and its arm about the hinge is z. Liquid 1 pushes the gate towards liquid 2 (normal towards liquid 1 is $-\vec{e}_x$), liquid 2 pushes it back.



Free-body diagram of the gate (unit width): the triangular gauge pressures $\rho_1 g(H_1 - z)$ and $\rho_2 g(H_2 - z)$, their resultants F_1 and F_2 at $H_1/3$ and $H_2/3$ above the hinge O, and the stop reaction R at the top

Moments about the Hinge

Moment of liquid 1 (turning the gate towards liquid 2):

$$M_1 = \int_0^{H_1} \rho_1 g(H_1 - z)\, z\, b\, dz = \rho_1 g b \left[\frac{H_1 z^2}{2} - \frac{z^3}{3}\right]_0^{H_1} = \rho_1 g b \left(\frac{H_1^3}{2} - \frac{H_1^3}{3}\right) = \frac{\rho_1 g b H_1^3}{6}$$

Moment of liquid 2 (turning it back), with the same integral and H_2 in place of H_1:

$$M_2 = \int_0^{H_2} \rho_2 g(H_2 - z)\, z\, b\, dz = \frac{\rho_2 g b H_2^3}{6}$$

Check with the resultants: each triangular distribution has the resultant $\rho_i g b H_i^2/2$ applied at its centroid, $H_i/3$ above the bottom, and $(\rho_i g b H_i^2/2)(H_i/3) = \rho_i g b H_i^3/6$.

Equilibrium and the Critical Level

Moment balance about the hinge, with the stop reaction R (pushing towards liquid 1) acting at the top, arm H_1:

$$M_1 - M_2 - R\,H_1 = 0 \quad \Longrightarrow \quad R = \frac{g b\,(\rho_1 H_1^3 - \rho_2 H_2^3)}{6 H_1}$$

The stop can only push, $R \geq 0$, so the configuration is in equilibrium while $\rho_1 H_1^3 \geq \rho_2 H_2^3$. The critical level is the one with $R = 0$:

$$\rho_1 H_1^3 = \rho_2 H_c^3 \quad \Longrightarrow \quad H_c = H_1 \left(\frac{\rho_1}{\rho_2} \right)^{1/3}$$

For $H_2 > H_c$ the moment of liquid 2 wins, the gate leaves the stop and turns towards liquid 1.

Answers and Checks

- $H_c = H_1 (\rho_1/\rho_2)^{1/3}$; equilibrium for $H_2 < H_c$; stop force $R = g b (\rho_1 H_1^3 - \rho_2 H_2^3)/(6 H_1)$.
- **Consistency with the statement**: $\rho_2 > \rho_1$ gives $H_c < H_1$, so the critical level lies within the gate, as it must for $H_2 < H_1$.
- **Limiting case** $\rho_2 = \rho_1$: $H_c = H_1$, two equal columns of the same liquid balance exactly.
- **Not the force balance**: the horizontal forces are equal when $\rho_1 H_1^2 = \rho_2 H_2^2$, which gives a different level, $H_1 (\rho_1/\rho_2)^{1/2}$; the gate turns about the hinge, so the moments decide (the difference is taken by the hinge reaction).
- **Dimensions**: H_c is H_1 times a dimensionless ratio, m; R is (m s^{-2})(m)(kg m^{-3} m^3)/m = N.
- **Check against the official key**: the sheet has no handwritten solution for this problem; the result was checked with SymPy (`tools/fluids_fs/verify_fluid_statics.py`).

PROBLEM FS6

Equilateral Triangle with Trapped Air

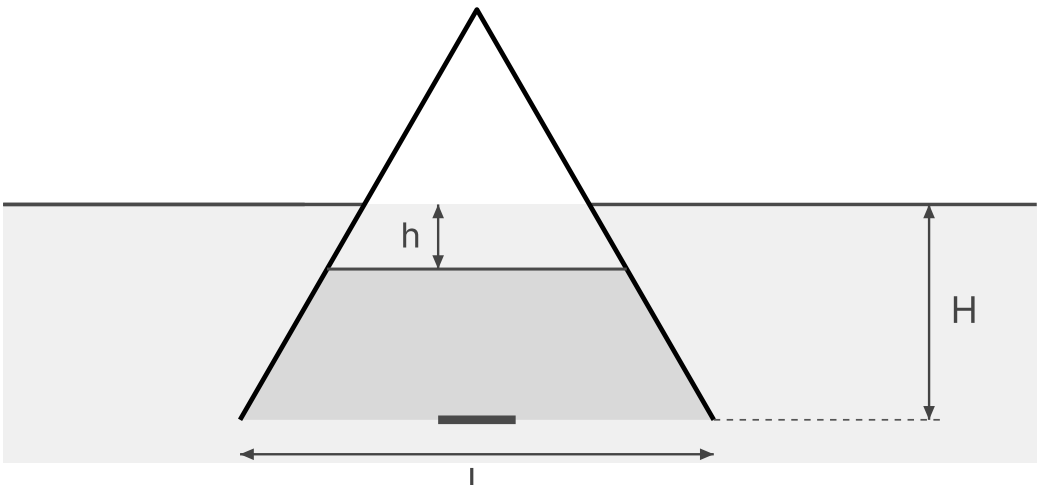
Original statement · fluid_statics_6.pdf

Floating body | Trapped air | Similar triangles | Erratum

OFFICIAL PROBLEM STATEMENT

An equilateral triangle of side L and weight W initially contains air at the ambient pressure and temperature, p_a and T_a. The triangle has an orifice in its lower side so that, when it is put into the water, the water enters it until the equilibrium position is reached. Obtain the system of equations that determines the equilibrium position, characterized by the heights H and h of the figure, and the pressure of the air inside the triangle. Assume that the sinking process is slow, so that the temperature of the gas stays constant at every moment.

Translated from the Spanish original. The triangle is a long prism (a two-dimensional problem): W and every area are per unit length normal to the paper.



Statement figure (redrawn from fluid_statics_6.pdf, p. 1): equilateral triangle of side L with an orifice in the lower side; its base is at the depth H below the outer free surface and the inner level is h below it

Source: fluid_statics_6.pdf (pages 1-2, sheet of 06-09-2008). Notes.pdf, Chapter 5: Eqs. 5.10, 5.15 and 5.18.

Worked solution

Hypotheses, Data and Unknowns

- Water and air at rest; water incompressible with density ρ; outside air at p_a.
- The trapped air has a uniform pressure p and keeps the temperature T_a (slow process): $p\,A = \text{const}$ per unit length, from (5.18) at constant mass and temperature.
- Thin walls; the weight of the air is negligible; the triangle floats with its base horizontal and its apex up (as drawn).
- Data: L, W (per unit length), ρ, g, p_a. Unknowns: H (depth of the base below the outer free surface), h (depth of the inner level below the outer free surface) and p.

Symbol	Meaning	SI unit
L	side of the triangle	m
W	weight per unit length	N/m
H, h	heights of the figure	m
p	final air pressure	Pa

Geometry: Similar Equilateral Triangles

The height of the equilateral triangle is $L\sqrt{3}/2$: in the right triangle formed by the height, half the base $L/2$ and a side L (hypotenuse), Pythagoras gives $\sqrt{L^2 - L^2/4} = L\sqrt{3}/2$.

Every horizontal cut of the triangle is the base of a smaller **equilateral** triangle with the same apex (same angles). The half-angle at the apex is 30°, so in the right triangle with the vertical height a (adjacent to 30°) and half the base (opposite), the half base is $a\tan 30^\circ = a/\sqrt{3}$. The area of a triangle of height a is then

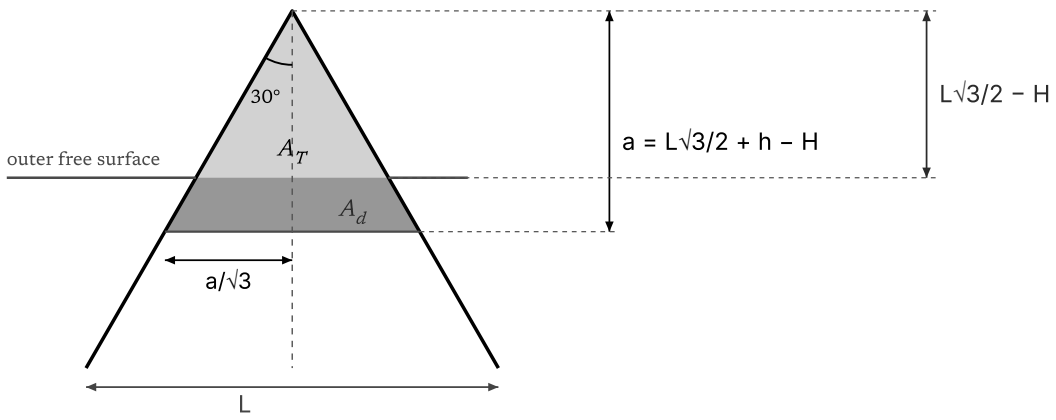
$$A(a) = \frac{1}{2}\cdot\frac{2a}{\sqrt{3}}\cdot a = \frac{a^2}{\sqrt{3}}$$

Check: $a = L\sqrt{3}/2$ gives $A = (3L^2/4)/\sqrt{3} = \sqrt{3}L^2/4$, the area A_0 of the whole triangle.

The base is at depth H and the inner level at depth h below the outer free surface, so the inner level is $H - h$ above the base, and the air triangle (from the inner level to the apex) has the height

$$a = \frac{L\sqrt{3}}{2} - (H - h) = \frac{L\sqrt{3}}{2} + h - H$$

The part of the triangle above the outer free surface has the height $L\sqrt{3}/2 - H$.



Every horizontal section of the equilateral triangle is the base of a smaller equilateral triangle with the same apex. The half-angle at the apex is $30°$, so a triangle of height a has the half base $a \tan 30° = a/\sqrt{3}$ and the area $a^2/\sqrt{3}$. Orange: the air triangle, height $a = L\sqrt{3}/2 + h - H$; green: the displaced area A_d between the outer free surface and the inner level; A_T: the part above the outer free surface

Equation 1: Isothermal Air

Initially the air fills the whole triangle, $A_0 = \sqrt{3}L^2/4$, at p_a; at equilibrium it fills the air triangle $A = a^2/\sqrt{3}$ at p. With constant mass and temperature, $p\,A = p_a A_0$:

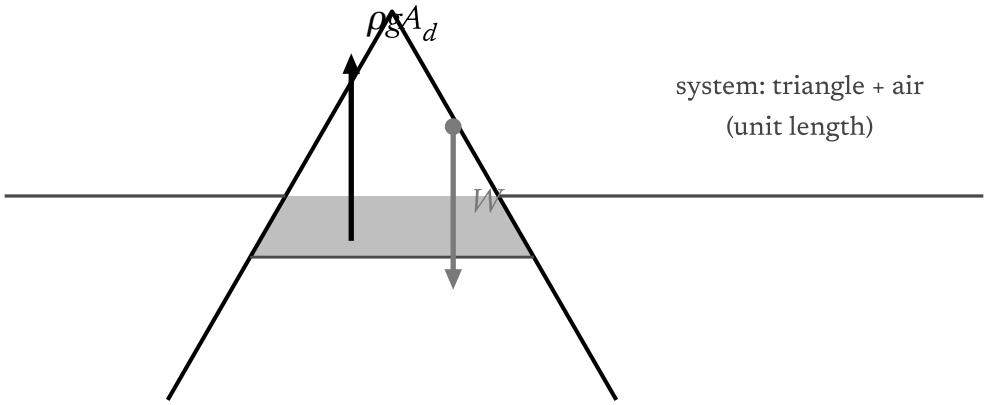
$$\frac{p}{p_a} = \frac{A_0}{A} = \frac{\sqrt{3}L^2/4}{a^2/\sqrt{3}} = \frac{3L^2}{4a^2} = \frac{3}{4}\,\frac{1}{\left(\frac{\sqrt{3}}{2} + \frac{h}{L} - \frac{H}{L}\right)^2}$$

Equation 2: the Hydrostatic Law

The water inside is connected with the outside through the orifice. From the outer free surface ($p = p_a$) down to the depth h of the inner level, (5.10) adds $\rho g h$, and at the interface the water pressure equals the air pressure:

$$p = p_a + \rho g h \quad \Longrightarrow \quad \frac{p}{p_a} = 1 + \frac{\rho g L}{p_a}\,\frac{h}{L}$$

Equation 3: Archimedes' Principle



Free-body diagram of the system triangle plus trapped air, per unit length: the weight W (air weight neglected) and the buoyancy $\rho g A_d$, where A_d is the part of the air region below the outer free surface (Archimedes, Eq. 5.15)

For the system triangle plus trapped air, the volume (area, per unit length) that displaces water is the air region below the outer free surface: the air triangle minus the triangle above the outer free surface,

$$A_d = \frac{a^2}{\sqrt{3}} - \frac{1}{\sqrt{3}}\left(\frac{L\sqrt{3}}{2} - H\right)^2$$

and the vertical equilibrium $W = \rho g A_d$ gives, dividing by $\rho g L^2$:

$$\frac{W}{\rho g L^2} = \frac{1}{\sqrt{3}} \left[\left(\frac{\sqrt{3}}{2} + \frac{h}{L} - \frac{H}{L} \right)^2 - \left(\frac{\sqrt{3}}{2} - \frac{H}{L} \right)^2 \right]$$

The same equation follows from the free-body diagram of the shell alone: the air pushes its inner walls with $p - p_a$ over the part above the inner level, the water pushes its outer walls below the outer level, and adding the vertical projections with the rule (5.14) gives $\rho g A_d$.

Answers and Checks

The equilibrium is the solution of the three equations in the three unknowns H/L, h/L, p/p_a, with the two dimensionless parameters $W/(\rho g L^2)$ and $\rho g L/p_a$:

$$\frac{p}{p_a} = \frac{3}{4\left(\frac{\sqrt{3}}{2} + \frac{h}{L} - \frac{H}{L}\right)^2}, \qquad \frac{p}{p_a} = 1 + \frac{\rho g L}{p_a}\,\frac{h}{L}, \qquad \frac{W}{\rho g L^2} = \frac{1}{\sqrt{3}} \left[\left(\frac{\sqrt{3}}{2} + \frac{h}{L} - \frac{H}{L} \right)^2 - \left(\frac{\sqrt{3}}{2} - \frac{H}{L} \right)^2 \right]$$

- **Numerical example** (solved with SymPy in `tools/fluids_fs/verify_fluid_statics.py`): for $\rho g L/p_a = 0.1$ and $W/(\rho g L^2) = 0.2$, $H/L = 0.244$, $h/L = 0.234$, $p/p_a = 1.023$.
- **Limiting case** $h = H = 0$ (triangle resting on the surface, no water inside): the first equation gives $p/p_a = (3/4)/(3/4) = 1$, as it must.
- **Expanding the bracket** of the third equation: $W/(\rho g L^2) = (h/L)\,[\sqrt{3} - 2H/L + h/L]/\sqrt{3}$, which vanishes for $h = 0$: with no compression there is no displaced air and no buoyancy.
- **Dimensions**: W is N/m and $\rho g L^2$ is (kg m^{-3})(m s^{-2})(m^2) = N/m; $\rho g L/p_a$ is Pa/Pa.

Erratum in the official solution

fluid_statics_6.pdf, page 2, writes $p/p_a = \frac{1}{4}(\sqrt{3}/2 + h/L - H/L)^{-2}$. The quotient of the areas is $(\sqrt{3}L^2/4) \cdot \sqrt{3}/a^2 = 3L^2/(4a^2)$, so the coefficient is $3/4$: with $h = H = 0$ the official formula gives $p = p_a/3$ instead of $p = p_a$. Equations 2 and 3 of the official key agree with the ones above.

PROBLEM FS7

Wedge between Two Liquids

Original statement · fluid_statics_7.pdf

Inclined surfaces | Two liquids | Force and moment | Virtual column

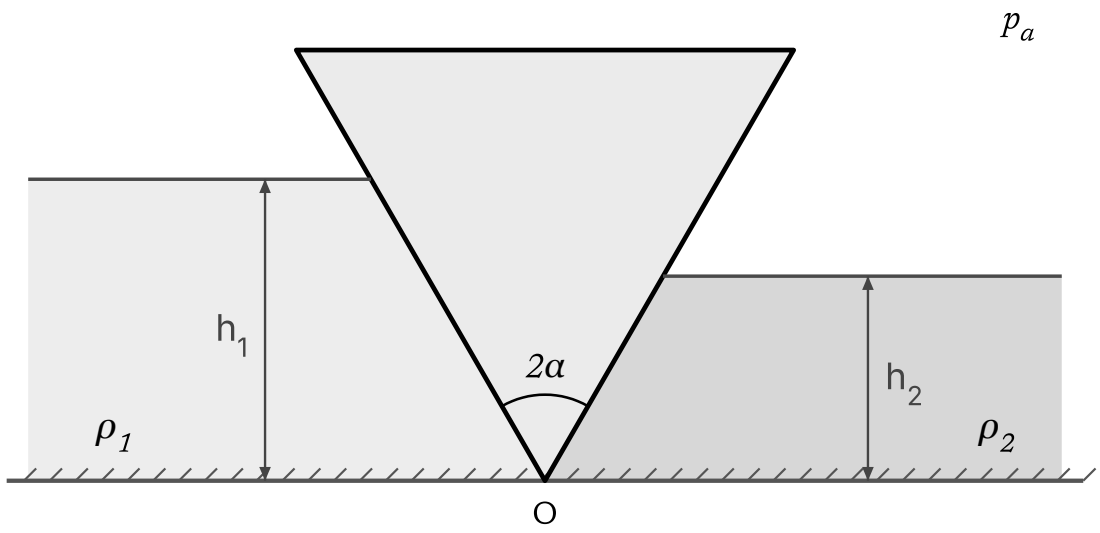
OFFICIAL PROBLEM STATEMENT

A two-dimensional wedge of angle 2α and weight W per unit length rests on its lower vertex in the position shown in the figure. The wedge separates two liquids of densities ρ_1 and $\rho_2 > \rho_1$. The levels of the two liquids are h_1 and h_2 respectively. Find:

1. The reaction force exerted on the wedge at the support point O.

2. The relation that must hold between h_1 and h_2 for the equilibrium position shown in the figure to be possible.

Translated from the Spanish original.



Statement figure (redrawn from fluid_statics_7.pdf, p. 1): wedge of angle 2α and weight W per unit length resting on its vertex O; liquid ρ_1 of depth h_1 on the left and $\rho_2 > \rho_1$ of depth h_2 on the right

Source: fluid_statics_7.pdf (pages 1-2, sheet of 05-09-07, problem P5). Notes.pdf, Chapter 5: Eqs. 5.10, 5.11 and 5.14.

Worked solution

Hypotheses, Data and Unknowns

- Both liquids at rest, incompressible; air at p_a above them and on the top face of the wedge, so the p_a contributions cancel over the closed surface and only gauge pressures count.
- Planar problem: forces and moments per unit length normal to the paper.
- The wedge is symmetric about the vertical through O, each face making the angle α with that vertical; its centre of gravity lies on that vertical, so W has no moment about O.
- At O the wedge rests on the floor; the reaction $\vec{R} = R_x\vec{e}_x + R_y\vec{e}_y$ acts at O (no moment at a point support).
- Data: α, W, ρ_1, ρ_2, h_1, h_2, g. Unknowns: R_x, R_y and the relation between h_1 and h_2.

Symbol	Meaning	SI unit
α	half-angle of the wedge	rad
W	weight per unit length	N/m
h_1, h_2	liquid levels above O	m
R_x, R_y	reaction at O per unit length	N/m

Coordinates, Element and Normal of Each Face

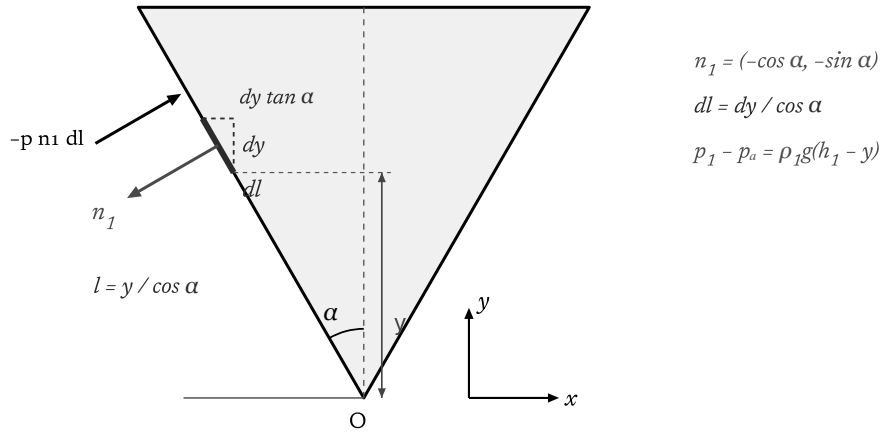
Origin at the vertex O, x horizontal to the right, y vertical and up. A point of the left face at the height y is at $x = -y\tan\alpha$: in the right triangle with the vertical leg y (adjacent to α) and the horizontal leg (opposite), the horizontal leg is $y\tan\alpha$, and the hypotenuse, the distance from O along the face, is $l = y/\cos\alpha$.

The element of the face between y and $y + dy$ is the hypotenuse of the small right triangle with the legs dy and $dy\tan\alpha$, so $dl = dy/\cos\alpha$.

The face makes the angle α with the vertical, so its unit normal makes the angle α with the horizontal. Liquid 1 is on the left and below the face, so the normal towards liquid 1 is

$$\vec{n}_1 = -\cos\alpha\,\vec{e}_x - \sin\alpha\,\vec{e}_y, \qquad \vec{n}_2 = \cos\alpha\,\vec{e}_x - \sin\alpha\,\vec{e}_y$$

($\vec{n}_2$ by mirror symmetry for the right face). Check: $\vec{n}_1$ is perpendicular to the face direction $(-\sin\alpha, \cos\alpha)$, since $\sin\alpha\cos\alpha - \sin\alpha\cos\alpha = 0$.



Left face of the wedge, making the angle α with the vertical. The element dl is the hypotenuse of the right triangle with the vertical leg dy (adjacent to α) and the horizontal leg $dy\tan\alpha$ (opposite), so $dl = dy/\cos\alpha$; for the same reason the distance from O along the face is $l = y/\cos\alpha$. The unit normal towards liquid 1 is $\vec{n}_1 = -\cos\alpha\,\vec{e}_x - \sin\alpha\,\vec{e}_y$ and the force on the wedge is $-p\vec{n}_1\,dl$

By (5.10), the gauge pressures are $p_1 - p_a = \rho_1 g(h_1 - y)$ for $0 \le y \le h_1$ and $p_2 - p_a = \rho_2 g(h_2 - y)$ for $0 \le y \le h_2$.

Forces on the Faces

On the left face, $d\vec{F}_1 = -(p_1 - p_a)\vec{n}_1\,dl = \rho_1 g(h_1 - y)(\cos\alpha\,\vec{e}_x + \sin\alpha\,\vec{e}_y)\,\dfrac{dy}{\cos\alpha}$, so

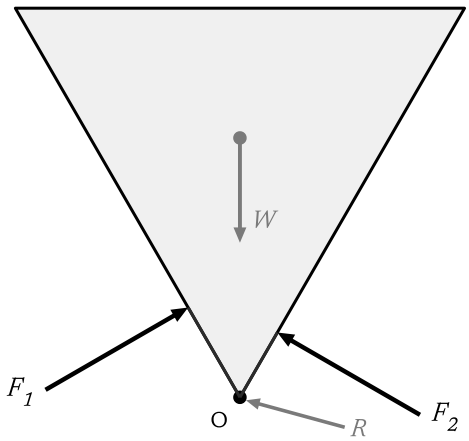
$$\vec{F}_1 = \rho_1 g\,(\vec{e}_x + \tan\alpha\,\vec{e}_y)\int_0^{h_1}(h_1 - y)\,dy = \rho_1 g\,(\vec{e}_x + \tan\alpha\,\vec{e}_y)\left[h_1 y - \frac{y^2}{2}\right]_0^{h_1} = \frac{\rho_1 g h_1^2}{2}\,(\vec{e}_x + \tan\alpha\,\vec{e}_y)$$

and in the same way on the right face, with $-\vec{n}_2 = -\cos\alpha\,\vec{e}_x + \sin\alpha\,\vec{e}_y$:

$$\vec{F}_2 = \frac{\rho_2 g h_2^2}{2}\,(-\vec{e}_x + \tan\alpha\,\vec{e}_y)$$

Check with the projection rule (5.14). The horizontal component of $\vec{F}_1$ is the force on the vertical projection of the face, a rectangle of height h_1 and unit width: $\rho_1 g h_1^2/2$. The vertical component is upward and equals the weight of the **virtual** column of liquid 1 above the face, the triangle with the vertical side h_1 and the horizontal side $h_1 \tan\alpha$: $\rho_1 g \cdot \frac{1}{2} h_1 \cdot h_1 \tan\alpha$. Both agree.

1. The Reaction at O



Free-body diagram of the wedge (per unit length): F_1 and F_2 normal to the faces, applied at the centres of pressure, at the distances $h_1/(3\cos\alpha)$ and $h_2/(3\cos\alpha)$ from O along the faces; the weight W on the symmetry line through O; the reaction R at O

Force balance of the wedge, $\vec{F}_1 + \vec{F}_2 - W\vec{e}_y + \vec{R} = 0$, component by component:

$$R_x = -\frac{\rho_1 g h_1^2}{2} + \frac{\rho_2 g h_2^2}{2}, \qquad R_y = W - \frac{\rho_1 g h_1^2}{2}\tan\alpha - \frac{\rho_2 g h_2^2}{2}\tan\alpha$$

2. The Moment about O and the Condition on the Levels

Every $d\vec{F}_1$ is normal to the face, so its arm about O is the distance along the face, $l = y/\cos\alpha$, and it turns the wedge clockwise (towards liquid 2). Its moment is

$$M_1 = \int_0^{h_1} (p_1 - p_a)\, l\, dl = \int_0^{h_1} \rho_1 g (h_1 - y)\, \frac{y}{\cos\alpha}\, \frac{dy}{\cos\alpha} = \frac{\rho_1 g}{\cos^2\alpha} \left[\frac{h_1 y^2}{2} - \frac{y^3}{3} \right]_0^{h_1} = \frac{\rho_1 g h_1^3}{6\cos^2\alpha}$$

The same follows from the cross product $\vec{r} \times d\vec{F}_1$ with $\vec{r} = (-y\tan\alpha, y)$: $M_z = x\, dF_y - y\, dF_x = -\rho_1 g (h_1 - y)\, y\, (\sin^2\alpha + \cos^2\alpha)\, dy/\cos^2\alpha$, clockwise, with $\sin^2\alpha + \cos^2\alpha = 1$. By symmetry, liquid 2 gives the counterclockwise moment $M_2 = \rho_2 g h_2^3/(6\cos^2\alpha)$. The weight and the reaction pass through O. Moment balance about O:

$$\frac{\rho_1 g h_1^3}{6\cos^2\alpha} = \frac{\rho_2 g h_2^3}{6\cos^2\alpha} \quad\Longrightarrow\quad \rho_1 h_1^3 = \rho_2 h_2^3 \quad\Longrightarrow\quad \frac{h_1}{h_2} = \left(\frac{\rho_2}{\rho_1}\right)^{1/3}$$

Answers and Checks

- $R_x = \frac{g}{2}(\rho_2 h_2^2 - \rho_1 h_1^2)$, $R_y = W - \frac{g}{2}(\rho_1 h_1^2 + \rho_2 h_2^2)\tan\alpha$.
- $h_1/h_2 = (\rho_2/\rho_1)^{1/3}$: the lighter liquid must stand higher ($h_1 > h_2$, as drawn). The angle α does not appear.
- **Support conditions (extension, not asked in the key)**: a vertex resting on the floor can only push up, so the position is possible only if $R_y \geq 0$, that is $W \geq \frac{g}{2}(\rho_1 h_1^2 + \rho_2 h_2^2)\tan\alpha$, and only if the support can supply the horizontal force R_x (friction or a notch at O).
- With the levels in that relation, $\rho_2 h_2^2 = \rho_1 h_1^2\,(h_1/h_2) > \rho_1 h_1^2$, so $R_x > 0$: the net push of the liquids is towards liquid 1 and the floor holds the vertex back.
- **Limiting case** $\rho_1 = \rho_2$: $h_1 = h_2$, $R_x = 0$ and $R_y = W - \rho g h^2 \tan\alpha$, the weight minus the buoyancy of the submerged triangle of area $h^2\tan\alpha$ (Archimedes' principle, with the vertex as the only support).
- **Dimensions**: $\rho g h^2$ is (kg m^{-3})(m s^{-2})(m^2) = N/m, like W; $\rho g h^3$ is N m/m.

- **Check against the official key**: fluid_statics_7.pdf gives $\vec{F}_1 = \rho_1 g h_1^2 (\vec{e}_x + \tan\alpha\, \vec{e}_y)/2$, $\vec{F}_2 = \rho_2 g h_2^2 (-\vec{e}_x + \tan\alpha\, \vec{e}_y)/2$, the same R_x and R_y, $M_1 = \rho_1 g h_1^3/(6\cos^2\alpha)$ and $h_1/h_2 = (\rho_2/\rho_1)^{1/3}$, the same results. The key writes the normals pointing into the wedge, the opposite of the convention of Eq. 5.11; the forces are the same.

PROBLEM FS9

Hinged Gate Opening Height

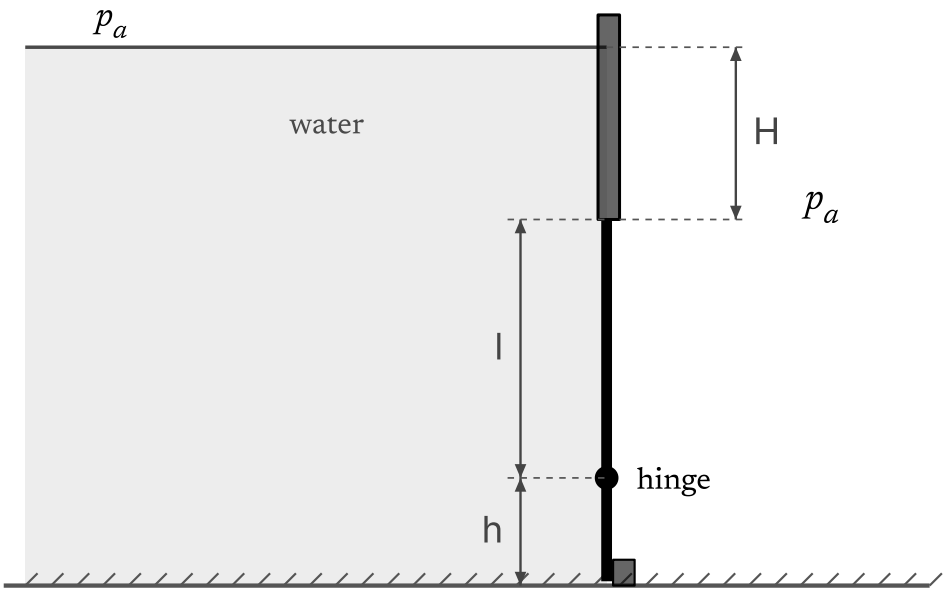
Original statement · fluid_statics_9.pdf

Plane surface | Moment about a hinge | Centre of pressure | Opening condition

OFFICIAL PROBLEM STATEMENT

Calculate the height H for which the hinged gate shown in the figure will open, if the hinge is at a distance l from the upper wall and at a known distance h from the floor ($h < l$).

Translated from the Spanish original. The upper wall is the fixed wall above the gate, so l is measured from the top edge of the gate.



Statement figure (redrawn from fluid_statics_9.pdf, p. 1): gate hinged at a distance l below its top edge and $h < l$ above the floor; the water surface is H above the top edge; air at p_a on both sides; a stop at the floor on the dry side

Source: fluid_statics_9.pdf (page 1, sheet of 10-09-04). Notes.pdf, Chapter 5: Eqs. 5.10, 5.11 and 5.13.

Worked solution

Hypotheses, Data and Unknowns

- Water at rest, incompressible, density ρ; air at p_a on the free surface and on the dry side, so only gauge pressures count.
- The gate is vertical, of height $l + h$ and width b, hinged without friction at a distance l below its top edge and h above the floor; its weight passes through the hinge and has no moment.
- A stop at the floor, on the dry side, keeps the lower edge from moving out: it can push the gate only towards the water.
- The free surface is at the height H above the top edge of the gate. Data: l, h, ρ, g, b. Unknown: the height H at which the gate opens.

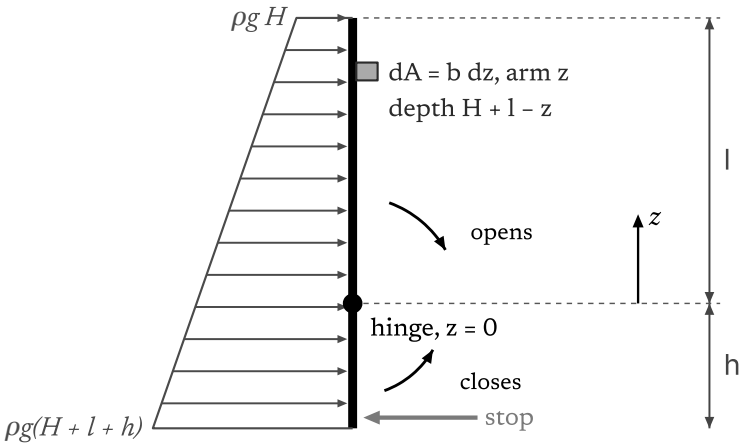
Symbol	Meaning	SI unit
H	water above the top edge of the gate	m
l, h	distances from the hinge to the top edge and to the floor	m
b	width of the gate	m

Coordinates and the Surface Element

Take z vertical, pointing up, with $z = 0$ at the hinge; the gate covers $-h \le z \le l$, and the free surface is at $z = H + l$. The depth of a point is $H + l - z$, so by (5.10)

$$p(z) - p_a = \rho g (H + l - z)$$

The strip between z and $z + dz$ has the area $b\,dz$ and the horizontal force $(p - p_a)\,b\,dz$ towards the dry side; its arm about the hinge is z. Strips above the hinge ($z > 0$) turn the gate with the top edge out (opening); strips below it ($z < 0$) turn it with the lower edge out, against the stop.



Free-body diagram of the gate (width b): the gauge pressure grows from $\rho g H$ at the top edge to $\rho g(H+l+h)$ at the floor; with z measured up from the hinge, the strip $b\,dz$ has depth $H+l-z$ and arm z. The part above the hinge turns the gate open; the part below presses it against the stop

Moment about the Hinge

Counting the opening sense as positive, the signed arm z takes care of both parts at once:

$$M = \int_{-h}^{l} \rho g(H+l-z)\, z\, b\, dz = \rho g b \left[(H+l)\frac{z^2}{2} - \frac{z^3}{3} \right]_{-h}^{l}$$

Barrow's rule, with $(-h)^2 = h^2$ and $(-h)^3 = -h^3$:

$$M = \rho g b \left[(H+l)\frac{l^2 - h^2}{2} - \frac{l^3 + h^3}{3} \right]$$

Opening Condition

While $M < 0$ the water presses the lower edge against the stop and the gate stays closed. It opens when M reaches zero and becomes positive; M grows with H because $l^2 - h^2 > 0$ ($h < l$). Setting $M = 0$:

$$(H + l)\frac{l^2 - h^2}{2} = \frac{l^3 + h^3}{3} \quad \Longrightarrow \quad H + l = \frac{2}{3}\frac{l^3 + h^3}{l^2 - h^2}$$

With $l^3 + h^3 = (l + h)(l^2 - lh + h^2)$ and $l^2 - h^2 = (l + h)(l - h)$:

$$H = \frac{2(l^2 - lh + h^2)}{3(l - h)} - l = \frac{2l^2 - 2lh + 2h^2 - 3l^2 + 3lh}{3(l - h)} = \frac{(l + h)(2h - l)}{3(l - h)}$$

The gate opens for any water level above this value.

Interpretation with the centre of pressure. $M = 0$ means that the resultant passes through the hinge: the centre of pressure is at the hinge. As H grows, the centre of pressure rises towards the centroid of the gate (Section 4.1 of Topic 5), which is $(l - h)/2$ above the hinge; the gate opens when the centre of pressure crosses the hinge from below.

Answers and Checks

- $H + l = \frac{2}{3}\frac{l^3 + h^3}{l^2 - h^2}$, that is $H = \frac{(l + h)(2h - l)}{3(l - h)}$.
- **When is it positive?** Only for $l/2 < h < l$. If $h \leq l/2$ the formula gives $H \leq 0$: even with the free surface at the top edge the centre of pressure is already above the hinge (for a triangular distribution it sits at a third of the height from the bottom, $(l + h)/3$ above the floor, which is above the hinge when $h < (l + h)/3$, i.e. $h < l/2$), so the gate cannot be kept closed. At $h = l/2$ the centre of pressure of the triangle is exactly at the hinge, $H = 0$.
- **Limiting case** $h \to l$ (hinge at the centre): $H \to \infty$; the centre of pressure is always below the centroid, so a gate hinged at its centre never opens (compare with FS4).
- **Dimensions**: H is m (cubic over quadratic lengths).

- **Check against the official key**: fluid_statics_9.pdf gives $M = \rho g[(H+l)(l^2-h^2)/2 - (l^3+h^3)/3]$ and $H + l = \frac{2}{3}(l^3+h^3)/(l^2-h^2)$, the same results.

PROBLEM FS10

Horizontal Force on a Cylinder

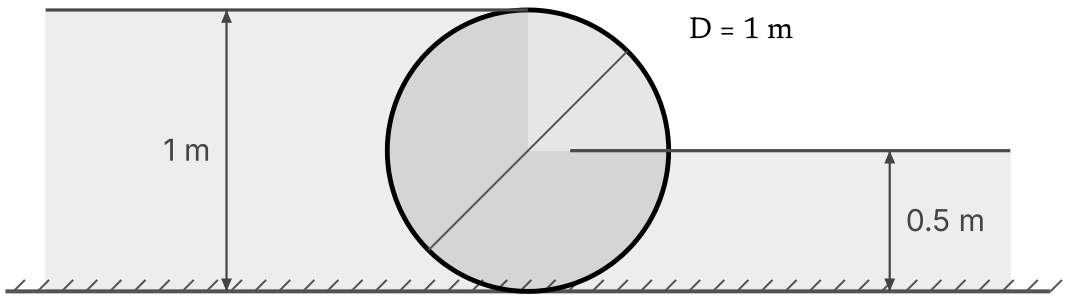
Original statement · fluid_statics_10.pdf

Curved surface | Polar coordinates | Projection rule | Simulator

OFFICIAL PROBLEM STATEMENT

Calculate the net horizontal force on the cylinder of length 5 metres shown in the attached figure.

Translated from the Spanish original.



cylinder length 5 m (normal to the paper)

Statement figure (redrawn from fluid_statics_10.pdf, p. 1): cylinder of diameter $D = 1$ m and length 5 m on the floor; water 1 m deep on the left, up to the top of the cylinder, and 0.5 m deep on the right, up to its centre

Source: fluid_statics_10.pdf (page 1, sheet of 12-9-01). Notes.pdf, Chapter 5: Eqs. 5.10, 5.11 and 5.14.

Worked solution

Hypotheses, Data and Unknowns

- Water at rest, incompressible, $\rho = 1000$ kg/m^3, $g = 9.81$ m/s^2; air at p_a above both free surfaces and on the dry parts of the cylinder, so only gauge pressures count.

- The cylinder, of diameter $D = 2R = 1$ m and length $L = 5$ m, lies on the floor with its axis horizontal and normal to the paper; end effects are neglected (the end faces are vertical, and their horizontal forces act along the axis, normal to the paper, so they do not enter the net force in the plane of the figure).
- Water on the left up to $2R = 1$ m (the top of the cylinder); on the right up to $R = 0.5$ m (its centre).
- Unknown: the net horizontal force F_x in the plane of the figure.

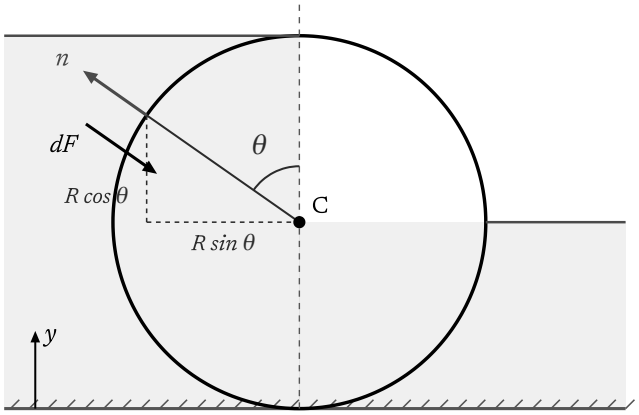
Symbol	Value	Meaning
R	0.5 m	radius
L	5 m	length
$2R$, R	1 m, 0.5 m	water depth on the left and on the right

Coordinates, Element and Normal

Take y vertical and up from the floor, x horizontal to the right, and measure the angle θ from the upward vertical through the centre C (at $y = R$). In the right triangle with the radius R as hypotenuse, the vertical leg $R\cos\theta$ is adjacent to θ and the horizontal leg $R\sin\theta$ opposite, so a point of the left half is at

$$x = -R\sin\theta, \qquad y = R + R\cos\theta, \qquad 0 \le \theta \le \pi$$

and a point of the right half at $x = +R\sin\theta$, same y. The element of the surface is the arc $R\,d\theta$ times the length L: $dS = LR\,d\theta$. The unit normal towards the water is the outward radial vector, with the horizontal component $n_x = -\sin\theta$ on the left half and $n_x = +\sin\theta$ on the right half.



y = R + R cos θ (from the floor)
depth on the left: 2R − y
n_x = −sin θ on the left half
n_x = +sin θ on the right half
dS = L R dθ

Element of the cylinder at the angle θ measured from the upward vertical through the centre C. In the right triangle with hypotenuse R, the horizontal leg $R\sin\theta$ is opposite to θ and the vertical leg $R\cos\theta$ adjacent, so the element is at $y = R + R\cos\theta$ above the floor. The unit normal towards the water on the left half is $\vec{n} = -\sin\theta\,\vec{e}_x + \cos\theta\,\vec{e}_y$ and the area is $dS = LR\,d\theta$

Left Half

The water on the left reaches $y = 2R$, so the depth of a point is $2R - y = R - R\cos\theta$ and, by (5.10), $p - p_a = \rho g R(1 - \cos\theta)$, over the whole left half ($0 \le \theta \le \pi$). The horizontal component of $-(p - p_a)\vec{n}\,dS$:

$$F_1 = -\int_0^{\pi} \rho g R(1-\cos\theta)(-\sin\theta)\,LR\,d\theta = \rho g R^2 L \int_0^{\pi} (\sin\theta - \sin\theta\cos\theta)\,d\theta$$

With $\int \sin\theta\,d\theta = -\cos\theta$ and $\int \sin\theta\cos\theta\,d\theta = \sin^2\theta/2$ (substitution $u = \sin\theta$, $du = \cos\theta\,d\theta$), by Barrow's rule:

$$F_1 = \rho g R^2 L \left[-\cos\theta - \frac{\sin^2\theta}{2}\right]_0^{\pi} = \rho g R^2 L\,[(1-0) - (-1-0)] = 2\rho g R^2 L$$

pointing to the right.

Right Half

The water on the right reaches $y = R$, so only the lower right quarter is wetted, $\pi/2 \le \theta \le \pi$, with the depth $R - y = -R\cos\theta$ (positive there, since $\cos\theta \le 0$):

$$F_2 = -\int_{\pi/2}^{\pi} \rho g(-R\cos\theta)(\sin\theta)\, LR\, d\theta = \rho g R^2 L \left[\frac{\sin^2\theta}{2}\right]_{\pi/2}^{\pi} = \rho g R^2 L \left(0 - \frac{1}{2}\right) = -\frac{\rho g R^2 L}{2}$$

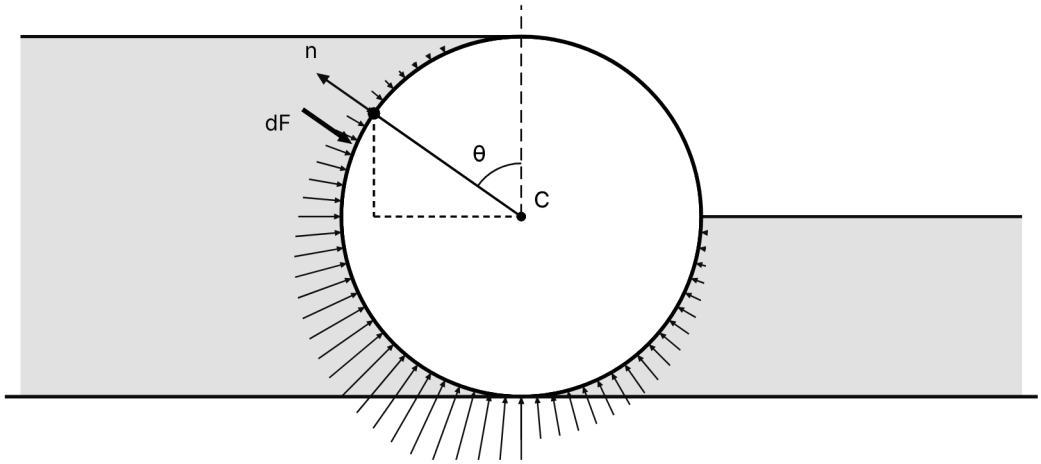
pointing to the left.

Check with the Projection Rule

By (5.14), the horizontal force on each side is the force on the vertical projection of the wetted surface. On the left, the projection is a rectangle of height $2R$ and length L with its centroid at the depth R: $\rho g R \cdot 2RL = 2\rho g R^2 L$. On the right, it is a rectangle of height R with its centroid at the depth $R/2$: $\rho g (R/2) \cdot RL = \rho g R^2 L/2$. Both agree with the integrals.

Simulator: the Curved Surface

`Simulator: cylinder, integration and projections`



Element: height y = R + R cosθ	`0.787 m`
Depth h_L − y, gauge pressure	`0.213 m, 2.09 kPa`
dF_x / dθ = (p − p_a) sinθ L R	`4.283 kN/rad (to the right)`
dF_y / dθ = −(p − p_a) cosθ L R	`-2.999 kN/rad (up positive)`
Left: F_1 = ρg L h_L^2/2	`24.525 kN, to the right`
Right: F_2 = ρg L h_R^2/2	`6.131 kN, to the left`
Net horizontal force F_1 − F_2	`18.394 kN (FS10: 18.394 kN)`

Answers and Checks

$$F_x = F_1 + F_2 = 2\rho g R^2 L - \frac{\rho g R^2 L}{2} = \frac{3}{2}\rho g R^2 L = \frac{3}{2}(1000)(9.81)(0.5)^2(5) = 18\,394\ \text{N} \approx 18.4\ \text{kN}$$

pointing to the right, from the deeper side to the shallower one.

- **Limiting case** equal levels on both sides: the two projections are equal and $F_x = 0$.
- **Dimensions**: $\rho g R^2 L$ is (kg m^{-3})(m s^{-2})(m^3) = N.

- **Beyond the statement**: the vertical force is upward and equals the weight of the water displaced by each wetted part (a half disc on the left, a quarter disc on the right); it is shown in the simulator but not asked for.
- **Check against the official key**: fluid_statics_10.pdf gives $F_1 = 2\rho g R^2 L$ and boxes $F_2 = -\rho g L R^2/2$ (the minus sign is the leftward sense, as above), then writes $F_T = F_1 - F_2 = \frac{3}{2}\rho g L R^2$ with F_2 taken as a magnitude; the net force is the same.